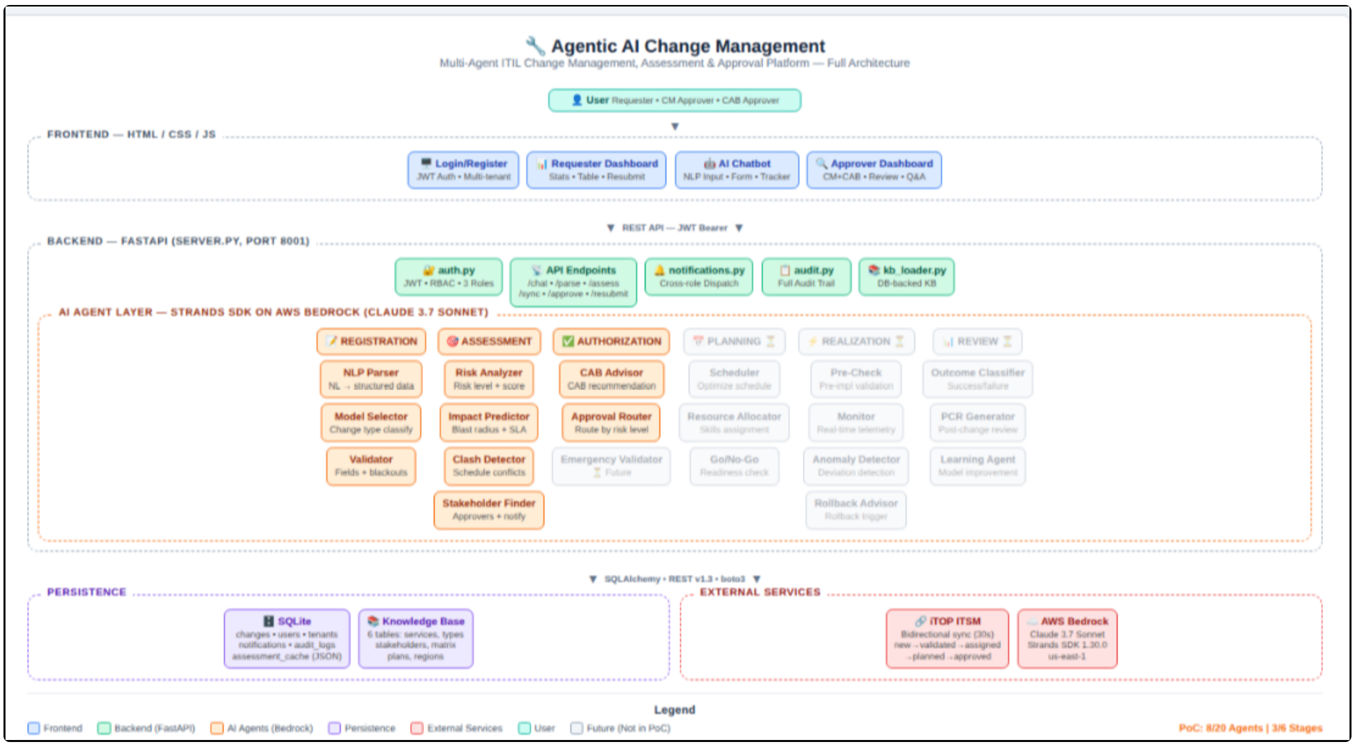
BT Change Management – Agentic AI

Scope

Agentic AI Change Management is an end-to-end automated ITIL change management platform for processing and governing infrastructure change requests across telecom networks. It covers NLP-driven change registration using a 3-agent Strands swarm on AWS Bedrock, automated risk and impact assessment via 4 specialized AI agents analyzing service dependencies, scheduling conflicts, and stakeholder impact, intelligent authorization routing with CAB advisory recommendations, bidirectional iTOP ITSM integration for full lifecycle management (creation through approval/rejection with resubmit on the same change), and 3 persona-based dashboards (Requester, CM Approver, CAB Approver) with contextual AI chatbot for approver Q&A. The platform supports a database-driven Knowledge Base covering telecom services, stakeholders, approval tiers, and change plan templates — with multi-tenant RBAC, background assessment caching for sub-second response times, editable AI outputs with cross-role notifications, and a complete audit trail for all decisions.

Architecture



Frontend — HTML / CSS / JavaScript
Login/Register page with JWT auth and multi-tenant support, Requester Dashboard with stat cards, change table, rejected filter and revise & resubmit flow, embedded AI Chatbot with NLP input, form sidebar and live status tracker, and a shared Approver Dashboard for CM and CAB personas with review modal, editable assessment blocks and contextual Q&A chatbot.

Backend — FastAPI (Python 3.10, Port 8001)
server.py as the main application with all REST endpoints, auth.py for JWT token management and RBAC with 3 roles (requester, cm_approver, cab_approver), notifications.py for cross-role notification dispatch, audit.py for full action audit trail, and kb_loader.py for database-backed Knowledge Base loading and seeding.

AI Agent Layer — Strands Agents SDK 1.30.0 on AWS Bedrock
9 agents implemented across 3 ITIL stages using Claude 3.7 Sonnet.

- Registration stage has a 3-agent swarm (NLP Parser, Model Selector, Validator) for natural language change extraction.
- Assessment stage runs 4 agents in background (Risk Analyzer, Impact Predictor, Clash Detector, Stakeholder Finder) with results cached in DB for instant retrieval.
- Authorization stage has a 2-agent swarm (CAB Advisor, Approval Router) for approval recommendations.

12 additional agents across Planning, Realization, and Review & Close stages are defined in the vision for future implementation.

Persistence / Storage
SQLite database (change_management.db) via SQLAlchemy ORM with tables for change requests, users, tenants, notifications, audit logs, and a JSON assessment_cache column for cached AI results.

Database-driven Knowledge Base with 6 tables covering services, change types, stakeholders (per-tenant), approval matrix, change plans, and regions.

External Services
iTOP ITSM server (localhost:8080) with bidirectional integration via REST API v1.3 — outbound for change creation and lifecycle stimuli (validate, assign, plan, approve), inbound via 30-second polling and webhooks for status sync.

AWS Bedrock (us-east-1) as the LLM provider for all AI agents.

Improving Application Performance: Latency Reduction, Robustness Hardening & Log Enrichment

1. Intelligent Validation & Follow-up Dialog
    a. Added rule-based Validation Agent (deterministic) that checks service, region, date, and change type against Knowledge Base
    b. Confidence-based routing: 0.9 auto-proceeds, 0.7–0.9 asks user to confirm, <0.7 generates follow-up questions with dropdowns
    c. Multi-turn conversation manager tracks session state and merges answers across turns
    d. Intelligent time slot generation with service criticality-aware maintenance windows, DB clash detection, blackout window checks, and scoring algorithm
    e. Follow-up Generator (deterministic) produces targeted drop down questions for missing fields.
2. Reliability & Latency Improvements
    a. Multi-layer timeout strategy: 8s Bedrock client timeout, 8s server-side thread timeout, 10s browser abort — prevents indefinite hangs
    b. Instant fallback parser (rule-based, <100ms) activates when Bedrock times out — detects service, region, change type via keyword matching and generates template-based plans
    c. Bedrock keep-alive mechanism if last Bedrock success was >5 minutes ago, uses fallback instantly.
3. Assessment orchestrator with Background Execution
    a. Assessment now triggers automatically on submit (background task) — approver sees results instantly when they open the review
    b. 4-agent sequential pipeline with per-agent progress tracking saved to DB (approver sees 1/4, 2/4... if they open during assessment)
    c. Optimized agent architecture: Risk Analyzer and Impact Predictor use Strands/Haiku (stochastic), Clash Detector uses DB rules + Haiku formatting (hybrid), Stakeholder Finder uses KB lookup + Haiku formatting (hybrid)
    d. Assessment latency reduced from 31s (Sonnet) to 14s (Haiku + tightened prompts)
    e. Prompt optimization: strict output templates, 100-150 word max, bullet-point-only format, max_tokens reduced from 2048 to 1024.
4. Approver Chat Improvements
    a. Switched from Sonnet to Haiku for approver Q&A — response time from 10-15s to 2-3s
    b. Concise conversational prompt: max 150 words, bullet points, specific to the change
    c. Clickable follow-up suggestion buttons after each response - approver can explore deeper with one click
    d. Markdown rendering in chat: tables, headers, bold, bullet points display properly
5. Observability - Execution Logs
    a. New execution_logs table tracks every pipeline step: agent name, status (success/slow/timeout/error/fallback), latency in ms, failure reason, user impact
    b. Logs captured for: NLP Parser, Validator, Confidence Router, Time Slot Generator, Submit, and all 4 assessment agents
    c. Each agent labeled with type: Stochastic (AI-driven), Deterministic (rules), Hybrid (rules + AI format)
    d. Per-change execution log visible in both Requester and Approver review modals (expandable section)
    e. System-wide execution log accessible via "Execution Logs" new item on both dashboards
    f. Enables instant debugging: "why was it slow?", "did anything fail?", "which agents ran?"

UC's

UC1 — **Register a change request via natural language**
A requester types a free-text description (e.g. "upgrade 5G core in Manchester this weekend"). The NLP Parser, Model Selector, and Validator agents extract structured fields — title, service, change type, target system, region — and suggest 3 scheduling slots with clash risk scores.

UC2 — Submit a change with instant iTOP creation
The requester confirms details and submits. The system creates the change in the local database and iTOP ITSM simultaneously, returns the change ID and iTOP reference in under 0.5 seconds, and queues background AI assessment.

UC3 — Automated risk and impact assessment
Four AI agents run asynchronously — Risk Analyzer produces a risk level and score, Impact Predictor maps blast radius and affected services, Clash Detector checks scheduling conflicts, and Stakeholder Finder identifies required approvers. Results are cached in the database for instant retrieval on review.

UC4 — CM Approver reviews and validates a change
The CM Approver opens the review panel showing full change details alongside AI assessment results. They can edit any assessment block (e.g. update stakeholder list), ask contextual questions via the AI chatbot, and validate the change on iTOP which advances it to CAB review.

UC5 — CAB Approver plans, schedules, and approves
The CAB Approver reviews the change with assessment context, pushes scheduled dates to iTOP via Plan & Schedule, and approves or rejects directly on iTOP. Status syncs back to the local system within 30 seconds and the requester is notified.

UC6 — Reject a change with reason visibility
When a change is rejected (by CM or CAB), the system determines the rejection source based on the lifecycle stage. The requester sees the change in the Rejected Requests section with a link to view the rejection reason in iTOP.

UC7 — Revise and resubmit a rejected change
The requester opens the rejected change, edits any field (justification, plan, schedule, etc.) via a pre-filled form, and resubmits. The system resets the same iTOP change back to "new" status, clears the old assessment, runs fresh AI agents, and notifies both CM and CAB approvers.

UC8 — Contextual Q&A for approvers
CM and CAB approvers can ask free-text questions about any change request in the review chatbot. The system builds context from the change details, cached assessment, and Knowledge Base, then calls Bedrock to provide ITIL-expert answers.

UC9 — Cross-role notification on assessment edits
When an approver edits an assessment field (e.g. stakeholder list), the system updates the cached assessment and notifies all other relevant parties — CM edits notify CAB + requester, CAB edits notify CM + requester.

UC10 — Multi-tenant registration with auto-provisioning
A new user registers with a tenant name and role. If the tenant is new, the system auto-creates default CM Approver and CAB Approver accounts, ensuring every tenant has a complete approval chain from the start.

Demo Link

Meeting with Sonal Karwal (EXT)-20260408_131856-Meeting Recording.mp4

ChangeManagementWithVoiceOver.mp4

Presentation link